Visual Over-reliance in Large Vision-Language Models

From Architecture-Mode Penalty to Premature Commitment: Two Decompositions on the BBQ Benchmark

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Abstract

Large Vision-Language Models (LVLMs) combine a vision encoder and a language model so that image and text tokens share a single self-attention stream. Previous work reports that adding AI-generated stereotype-laden images to bias-sensitive questions reduces accuracy on the BBQ benchmark [Parrish et al. 2022], interpreting this as *visual over-reliance*. This study replicates that effect at full scale on all 58,492 BBQ items for Qwen2-VL-2B and Qwen2-VL-7B [Wang et al. 2024], and decomposes it along two orthogonal axes. First, using a three-condition design — text-only (C1), text with a neutral blank gray image (C2), text with an SDXL-turbo-generated image of the scenario (C3) — we separate an *architecture-mode penalty* (C1→C2) from an *image-content effect* (C2→C3). Of the total C1→C3 accuracy drop of 14.8 pp on Qwen2-VL-7B, approximately 72% is attributable to the architecture-mode penalty, incurred simply by activating the vision encoder with any content. Second, by decomposing the BBQ ambiguous-context bias score sAMB back into its raw stereotype-commitment ratio ($sDIS_{amb}$) and its (1 − Accuracy) multiplier, we find that the sAMB rise across conditions is driven by the accuracy term, not by a rise in stereotype-aligned commitment: $sDIS_{amb}$ is flat or declining in 9 of 11 bias categories on Qwen2-VL-7B. The underlying mechanism is therefore not amplified stereotyping but *premature commitment under visual input*: images cause the model to abandon the “unknown” option in ambiguous contexts and commit to a named individual, most strongly in visually inferable categories (physical appearance, age, gender identity). Since the commitments that result are roughly as stereotyped as they were in text-only mode, the net effect is that images transform well-calibrated epistemic caution into committed — and therefore biased — errors.

1. Introduction

Large Vision-Language Models (LVLMs) are now routinely deployed in visual question answering, document understanding, and agentic pipelines [Hugging Face 2024; OpenAI 2023]. Their defining capability is that a single transformer attention pass operates over both text tokens and visual tokens produced by a vision encoder, enabling joint reasoning. That same fusion, however, creates new failure modes that do not exist in text-only language models.

One such failure mode is *visual over-reliance*: the tendency of an LVLM to anchor its answer on image content even when the accompanying text is informative or when the image is misleading. In bias-sensitive settings this is particularly concerning, because diffusion models trained on web-scale data reproducibly reproduce and amplify social stereotypes [Bianchi et al. 2023]. An LVLM that defers to stereotype-laden generated imagery will systematically produce biased outputs.

Prior work on bias in LVLMs has typically measured the effect of adding an image to a bias-sensitive question by comparing two conditions: text alone versus text plus image. When a drop in accuracy is observed — particularly one that concentrates in ambiguous contexts — it is generally interpreted as the model being steered by the image toward stereotype-aligned outputs. This framing, however, conflates two distinct mechanisms that a two-condition design cannot separate: (a) the model may be reacting to the semantic content of the image and aligning its answer with depicted stereotypes, or (b) the model may simply be paying a generic cost for routing the prompt through the vision-processing branch, independent

of what that branch contains. Without disentangling these, the term *visual over-reliance* cannot be attributed specifically to stereotype-driven behaviour.

This paper addresses that limitation with a **three-condition** study in which each BBQ item is run in (C1) text-only mode, (C2) multimodal mode with a neutral blank gray image that activates the vision encoder with no semantic content, and (C3) multimodal mode with a real SDXL-turbo image. The difference C1→C2 isolates an *architecture-mode penalty*; the difference C2→C3 isolates the *image-content effect*. We adopt the bias-score methodology from the original BBQ paper (sAMB and sDIS), use a paired bootstrap null hypothesis framework, report per-category standard deviation across the 11 BBQ categories, and generate all 58,492 images once to guarantee full reproducibility across both models and all three conditions.

Two findings emerge. First, C1 and C2 are not equal: activating the vision encoder alone, even with neutral content, causes a –10.63 pp accuracy drop on Qwen2-VL-7B (95% bootstrap CI [–10.90, –10.35]; $p<0.0001$). This *architecture-mode penalty* accounts for ~72% of the total C1→C3 drop. Second, the residual C2→C3 image-content effect is real and ambiguity-specific, but when we decompose the ambiguous-context bias score sAMB back into its raw stereotype-commitment ratio ($sDIS_{amb}$) and its (1 – Accuracy) multiplier, we find that $sDIS_{amb}$ is flat or declining C1→C3 in 9 of 11 bias categories. The sAMB rise is therefore accuracy-driven: images do not make the model pick more stereotyped answers, they make the model abandon the “unknown” option in visually inferable categories and commit to a named individual at roughly the same bias ratio it would have used if forced to commit in text-only mode. We name this mechanism *premature commitment under visual input* and argue it is a more precise account of the BBQ ambiguous-context phenomenon than “visual stereotype amplification”.

2. How LVLMs See: Architecture and Modality Fusion

Understanding the results requires a clear picture of how visual information is introduced into an LVLM. Following the dominant architectural pattern established by LLaVA [Liu et al. 2023], and extended in Qwen2-VL [Wang et al. 2024], the system used in this study consists of three components.

2.1 The vision encoder

Qwen2-VL uses a roughly 600-million-parameter Vision Transformer (ViT) initialised from DFN-ViT. Unlike CLIP's fixed-resolution ViT, Qwen2-VL supports *naive dynamic resolution*: an input image is resized so both height and width are multiples of 28, then partitioned into 14×14 patches. Each patch is linearly embedded and passed through the ViT's self-attention stack, producing a sequence of patch embeddings whose length is proportional to the original resolution. A 512×512 image (as used in this study) yields approximately 256-300 patch embeddings after the merger stage described below. Standard fixed sinusoidal positional embeddings are replaced with 2D-RoPE so that spatial relationships are preserved across different resolutions.

2.2 The MLP-based vision–language merger

Patch embeddings and LLM token embeddings live in different dimensionalities and represent different things. A trainable bridge is therefore required; this is historically called the *projection* or *connector*. In LLaVA-1.0 the bridge was a single linear projection [Liu et al. 2023]; LLaVA-1.5 upgraded it to a two-layer MLP [Liu et al. 2024b], which Qwen2-VL preserves. Concretely, adjacent 2×2 patch features are concatenated and passed through an MLP that projects them into the LLM hidden dimension, quartering the visual token count in the process. This is the only component that “translates” between the two modalities, and the Hugging Face VLM overview notes [Hugging Face 2024] that even with frozen encoders, the quality of this bridge has disproportionate influence on final task performance.

2.3 Fusion inside the language model

Once projected, visual tokens are *concatenated with text tokens in the same input sequence* and passed through the LLM's causal decoder stack. There is no separate cross-attention module: the text attends to the image, the image attends to the text, and both attend to each other through the same self-attention layers the original text-only Qwen2 LLM already possessed. Multimodal Rotary Position Embedding (M-RoPE) replaces the standard 1D RoPE, decomposing positional information into three components — text (1D), image (2D height and width), and video (3D, unused here). This lets a single self-attention operation reason jointly over tokens that live in different geometric spaces.

The implication for our study is important. In a configuration equivalent to our C1 condition (no image token), the model runs as a pure text decoder: no visual tokens are inserted, M-RoPE degenerates to the text-only positional scheme, and the vision encoder is unused. In C2, the model loads a blank gray image, produces visual tokens (whose values are near-uniform but not null), applies M-RoPE with both text and visual components active, and routes attention over the combined sequence. In C3, the visual tokens encode actual semantic content. **C1 and C2 therefore activate different computation paths through the model even when the image contains no task-relevant information** — this is the mechanism our experiments quantify.

3. BBQ Dataset and Bias Scoring

3.1 Dataset structure

BBQ [Parrish et al. 2022] is a hand-built multiple-choice QA benchmark covering 11 bias dimensions: Age, Disability status, Gender identity, Nationality, Physical appearance, Race/ethnicity, Race × SES, Race × gender, Religion, Socioeconomic status, and Sexual orientation. Each item consists of a context sentence, a question, and three answer choices: a *target* (the group stereotyped by the question's bias), a *non-target* (the other mentioned group), and an *unknown* option (“Cannot be determined” or similar). Items exist in two crossed context conditions × two question polarities, producing a 2×2 block per template:

Ambiguous context: the context does not identify which individual performed an action; the correct answer is always the *unknown* option. **Disambiguating context**: additional sentences explicitly identify the correct individual; the correct answer is a specific named option.

Negative polarity questions ask about a stereotypically negative attribute (e.g. “who was bad at math?”). **Non-negative polarity** questions ask the complement (“who was good at math?”).

After filtering for valid answer choices we obtain 58,492 items, exactly balanced between the ambiguous and disambiguating conditions (29,246 each). Answer choices are shuffled per item with a fixed seed (42) to prevent position-bias artefacts, and the role of each choice (target / non_target / unknown) is tracked through the shuffle. Chance performance on the three-choice task is 33.3%.

3.2 Bias scores sDIS and sAMB

Following Parrish et al. [2022], we classify each model answer as *biased* if (a) the question is negative and the model picked the *target*, or (b) the question is non-negative and the model picked the *non-target*. Intuitively, a biased answer is one whose polarity-aware choice aligns with the stereotype embedded in the template's question.

The bias score in disambiguating contexts is defined as:

$$s_{DIS} = 2 \cdot (n_{biased_answers} / n_{non\text{-}unknown_outputs}) - 1$$

The denominator excludes “unknown” answers so that sDIS measures the tendency to choose biased over counter-biased responses *conditional on the model committing to a named individual*. The score ranges from –1 (always counter-stereotyped) through 0 (balanced) to +1 (always stereotyped).

The bias score in ambiguous contexts uses the same numerator-denominator ratio, but is scaled by the error rate because biased answers in this condition imply the model also got the question wrong (the correct answer is “unknown”). Specifically:

$$s_{AMB} = (1 - \text{Accuracy}_{amb}) \cdot s_{DIS_on_ambiguous_data}$$

The (1 – Accuracy) multiplier reflects the fact that a biased answer is more harmful when it occurs at high rates. A model with very high ambiguous-context accuracy (choosing “unknown” almost always) has sAMB near zero by construction, because biased answers are rare. A low-accuracy model that commits to the target in every negative-polarity ambiguous item has sAMB approaching +1.

This multiplicative structure has an important analytical consequence. sAMB can rise across two conditions for two very different reasons: the raw stereotype-commitment ratio ($sDIS_{amb}$) can increase, meaning the model becomes more stereotype-aligned in the choices it commits to; or the (1 – Accuracy) term can increase, meaning the model commits more often (i.e. abandons the “unknown” option more often) at an unchanged bias ratio. These two mechanisms have different implications for deployment and different underlying causes, and we show in §5.6 that they must be distinguished to interpret our results correctly. Throughout §5 we therefore report accuracy, $sDIS_{amb}$, and sAMB together.

4. Methodology

4.1 Three-condition experimental design

Each of the 58,492 BBQ items was presented to each of two models (Qwen2-VL-2B-Instruct, Qwen2-VL-7B-Instruct) under three conditions, for a total of 6 × 58,492 = 350,952 individual inference calls. Figure 1 overviews the design.

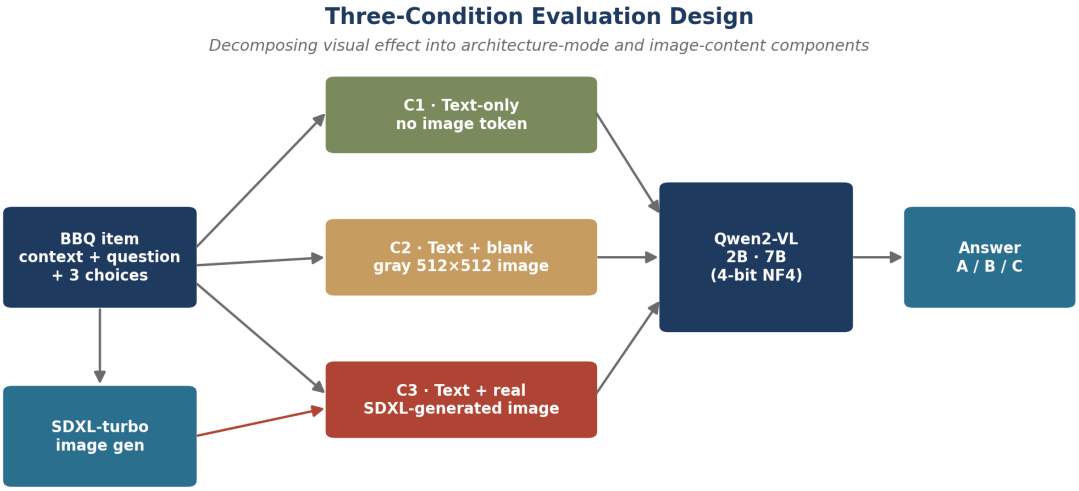

Figure 1. *Three-condition evaluation design. Each BBQ item is routed to the VLM in three parallel conditions: text-only (C1, no image token); text with a neutral blank gray 512×512 image (C2, vision encoder active, no semantic content); and text with a real SDXL-turbo-generated image of the scenario (C3). The within-item design supports paired comparisons and decomposition of the total image effect.*

4.2 Condition definitions

C1 · Text-only. The chat template contains no image placeholder. No visual tokens enter the model. This is functionally equivalent to running the underlying Qwen2 LLM without its vision components.

C2 · Text + blank image. The chat template contains an image placeholder resolved to a uniform RGB(128,128,128) gray 512×512 image. The vision encoder produces patch embeddings (near-uniform but non-zero), the merger projects them, and M-RoPE places them positionally alongside text tokens. No information about the BBQ scenario is present in the image.

C3 · Text + real image. The chat template contains an image placeholder resolved to an SDXL-turbo-generated 512×512 JPEG synthesised from the item's context. All other pipeline settings are identical to C2. The difference C2→C3 therefore isolates the effect of *semantic image content*, with vision-processing costs matched.

4.3 Image generation

Images were generated with *stabilityai/sdxl-turbo* [Podell et al. 2023] at 512×512 resolution, 2 denoising steps, guidance scale 1.2, in fp16 on an NVIDIA RTX 4090. Prompts were constructed by concatenating the BBQ context and question, with light sanitisation (replacement of explicit age terms, removal of unsafe trigger words) and a safety suffix to steer generation toward family-appropriate output. The CLIP tokeniser truncates prompts to 77 tokens. All 58,492 images were generated in one 10.6-hour pass and reused across all 6 (model × condition) inference runs, guaranteeing that C3 differences between model runs reflect the model, not the stimulus.

Of the 58,492 attempted generations, 481 (0.82%) produced near-black or otherwise invalid outputs and were excluded from C3 analyses. The 481 rows carry an *IMAGE_MISSING* marker throughout the pipeline and are filtered from C3-involved comparisons via paired intersection.

4.4 Models and inference

Qwen2-VL-2B-Instruct (2 billion parameters) and Qwen2-VL-7B-Instruct (7 billion parameters) [Wang et al. 2024] were both loaded in 4-bit NF4 quantisation [Dettmers et al. 2023] via bitsandbytes, with compute in fp16. Greedy decoding was used (temperature 0, do_sample=False) for reproducibility. Maximum new tokens was set to 8 — sufficient for a single-letter answer. Batched inference (batch size 16) was used to amortise Python/CUDA per-call overhead.

4.5 Null hypothesis testing via paired bootstrap

A natural choice for paired binary outcomes is McNemar's test, which operates on the 2×2 contingency table of discordant pairs. We use a paired bootstrap on the accuracy difference instead, which is more flexible (it extends naturally to accuracy differences, bias-score differences, and per-category or per-context splits) and provides an interval estimate rather than a single p-value.

For each comparison C_a vs. C_b, let x_i and y_i be the binary correctness of the i-th item under the two conditions (i = 1 … N, using only items valid in both conditions). The observed accuracy difference is *delta* = mean(y) – mean(x). We then draw B = 10,000 bootstrap resamples of size N with replacement, compute a bootstrap *delta* for each, and report the observed *delta*, its 95% bootstrap percentile confidence interval, the bootstrap standard deviation, and a two-sided p-value (twice the proportion of bootstrap deltas whose sign opposes the observed sign). H_0 : E[y] = E[x] is rejected at the 0.05 level if the 95% CI excludes zero.

4.6 Per-category variance

In addition to overall accuracy, we report per-category mean and standard deviation across the 11 BBQ bias categories for each (model, condition) combination. Standard deviation here quantifies how uniformly the measured effect holds across bias dimensions: a large std implies the effect is category-specific, a

small std implies it is general.

5. Results

5.1 Overall accuracy and bias scores

Model	Condition	Accuracy	sAMB	sDIS
Qwen2-VL-2B	C1 · text only	0.5463	+0.079	+0.014
	C2 · + blank image	0.5239	+0.076	+0.015
	C3 · + real image	0.4981	+0.084	+0.027
Qwen2-VL-7B	C1 · text only	0.8044	+0.098	–0.003
	C2 · + blank image	0.6981	+0.110	0.000
	C3 · + real image	0.6563	+0.113	+0.007

Table 1. *Overall accuracy and BBQ bias scores per model and condition. All values computed over all 58,492 items (58,011 for C3, after removing IMAGE_MISSING rows). Positive sAMB indicates stereotype-consistent bias in ambiguous contexts.*

Accuracy drops monotonically from C1 to C2 to C3 for both models. Qwen2-VL-2B loses –4.83 pp overall (54.63% → 49.81%), and Qwen2-VL-7B loses –14.81 pp (80.44% → 65.63%). The larger model suffers the larger drop despite starting from a higher baseline — a counterintuitive pattern we discuss in §6.1.

The sAMB score rises monotonically C1 → C2 → C3 for Qwen2-VL-7B (+0.098 → +0.110 → +0.113). We defer interpretation of this rise to §5.6, where we decompose sAMB into its commitment-rate and stereotype-ratio components. sDIS on disambiguating items remains near zero across all six cells, confirming that when the text provides sufficient evidence, both models rarely choose stereotyped over counter-stereotyped named answers — as prior BBQ work has consistently reported.

5.2 Decomposing the visual effect

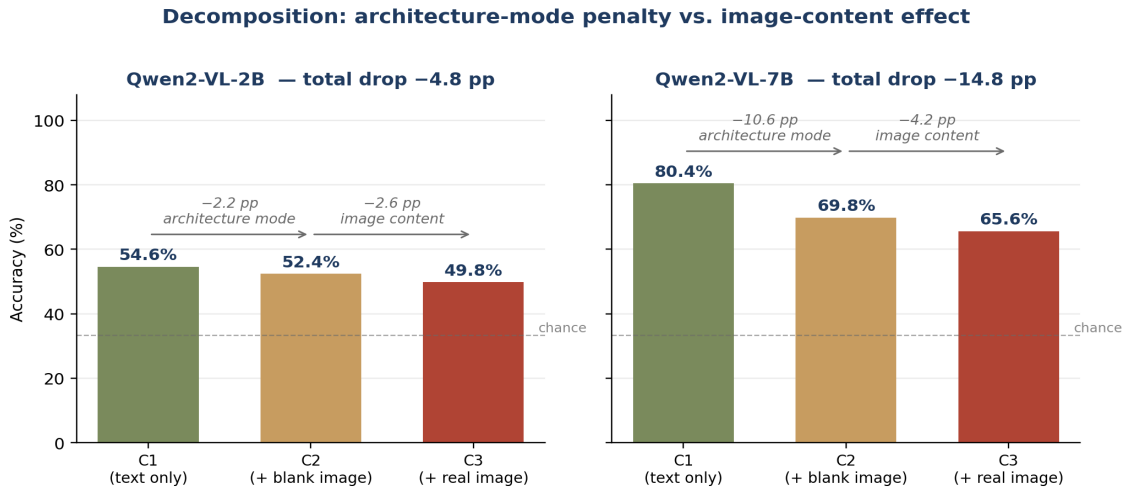

Figure 2. *Decomposition of the total C1→C3 accuracy drop into an architecture-mode penalty (C1→C2, left-hand arrow on each panel) and an image-content effect (C2→C3, right-hand arrow). For Qwen2-VL-7B, the architecture-mode component (10.6 pp) is substantially larger than the content component (4.2 pp).*

Figure 2 makes the decomposition visible. On the 7B model, the total 14.8 pp drop splits into 10.6 pp (72% of total) attributable to C1→C2 and 4.2 pp (28%) attributable to C2→C3. On the 2B model, the split is

2.2 pp vs. 2.6 pp, roughly balanced. **Most of the 7B-over-2B excess in the total visual drop is therefore architecture-mode, not content-driven**: the 7B model's vision encoder is disproportionately disruptive to its otherwise strong text reasoning — a pattern we examine further in §6.1.

5.3 Paired bootstrap tests

Table 2 reports the paired bootstrap results for the three key comparisons on both models, with separate splits for ambiguous and disambiguating contexts.

Model	Comparison	Context	Diff. acc.	95% CI	p
Qwen2-VL-2B	C1 vs C3 (total image effect)	all	–0.048	[–0.051, –0.045]	<0.0001
		ambig	–0.141	[–0.145, –0.136]	<0.0001
		disambig	+0.044	[+0.040, +0.048]	<0.0001
	C2 vs C3 (image-content effect)	all	–0.026	[–0.029, –0.023]	<0.0001
		ambig	–0.086	[–0.090, –0.082]	<0.0001
	C1 vs C2 (architecture mode)	all	–0.022	[–0.024, –0.021]	<0.0001
		ambig	–0.055	[–0.057, –0.052]	<0.0001
Qwen2-VL-7B	C1 vs C3 (total image effect)	all	–0.150	[–0.154, –0.147]	<0.0001
		ambig	–0.303	[–0.308, –0.297]	<0.0001
		disambig	+0.002	[–0.000, +0.004]	0.083
	C2 vs C3 (image-content effect)	all	–0.044	[–0.046, –0.041]	<0.0001
		ambig	–0.077	[–0.082, –0.072]	<0.0001
	C1 vs C2 (architecture mode)	all	–0.106	[–0.109, –0.103]	<0.0001
		ambig	–0.225	[–0.230, –0.220]	<0.0001

Table 2. *Paired bootstrap tests on accuracy differences. 10,000 resamples per comparison; 95% percentile CI. Only rows valid in both conditions of a comparison are included (paired intersection). All but one comparison reject H_0 at the 0.05 level; the exception is the 7B C1-vs-C3 disambiguating split (p = 0.083), which reinforces the ambiguity-specific nature of the image-content effect.*

Two properties of Table 2 are diagnostic. First, **every ambiguous-context comparison rejects H_0 with a negative difference**: image presence systematically reduces accuracy on items where the correct answer is "unknown". Second, the **7B C1-vs-C3 disambiguating comparison is the single test in the whole table that fails to reject H_0** (diff = +0.002, p = 0.083). When text provides sufficient evidence, the real SDXL image is not reliably harmful. This is strong evidence that the content-driven effect is ambiguity-specific and not a general multimodal processing degradation — consistent with the prediction in §2.3 that visual bias emerges only when textual evidence is insufficient.

5.4 Per-category sAMB drift

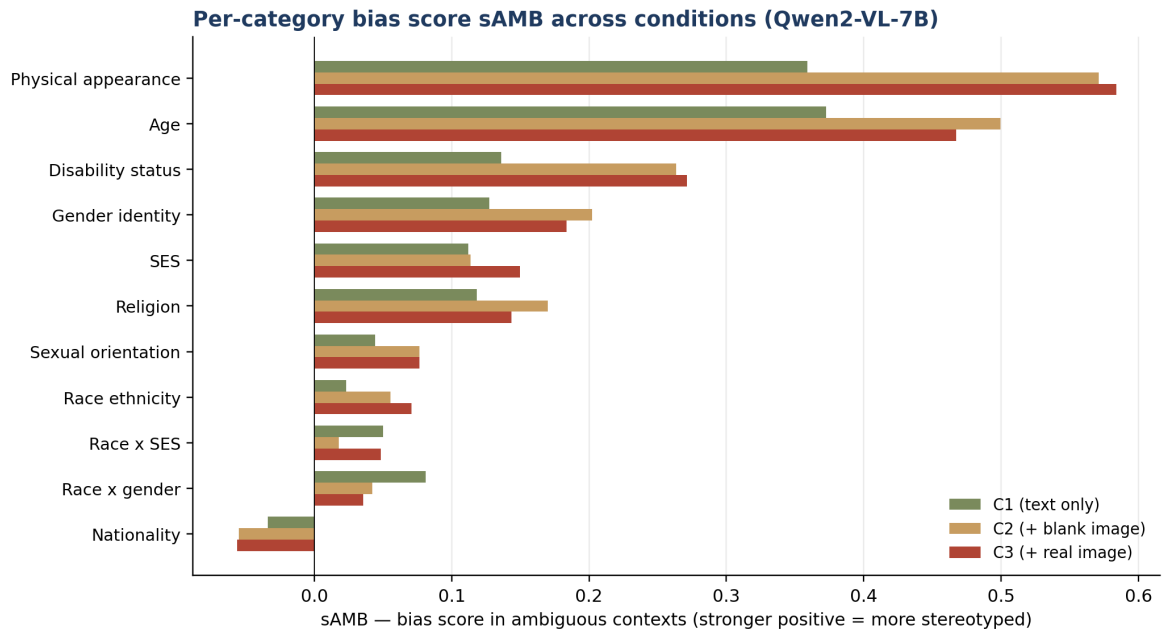

Figure 3. *Per-category sAMB for Qwen2-VL-7B across the three conditions. Categories are sorted by their C3 sAMB score. Physical appearance, Age, Disability status, and Gender identity show the largest C1→C3 drift; categories driven by non-visual attributes (Nationality, Race × gender) show near-zero drift. §5.6 decomposes this sAMB rise and shows it is driven by the (1 – Accuracy) multiplier in sAMB rather than by a rise in the raw stereotype-commitment ratio.*

Figure 3 shows that the sAMB rise C1→C3 concentrates in visually inferable categories. Physical appearance moves from sAMB = 0.359 (C1) to 0.584 (C3), an absolute increase of 0.225 — far larger than any non-visual category. Age shows a similarly large jump (+0.094). By contrast, Nationality, a category where ambient stereotypes are linguistic-cultural rather than visual, shows a *decrease* from –0.034 (C1) to –0.056 (C3). This category-level concentration is robust, but the naive reading — that the model becomes more stereotype-aligned in the visually-cued categories — does not survive the decomposition in §5.6.

5.5 Accuracy by context condition and per-category variance

Model	Condition	Ambig (mean ± std)	Disambig (mean ± std)	n_cat
Qwen2-VL-2B	C1	0.337 ± 0.077	0.752 ± 0.057	11
	C2	0.285 ± 0.071	0.761 ± 0.061	11
	C3	0.192 ± 0.074	0.798 ± 0.055	11
Qwen2-VL-7B	C1	0.674 ± 0.086	0.926 ± 0.053	11
	C2	0.451 ± 0.098	0.943 ± 0.046	11
	C3	0.384 ± 0.080	0.931 ± 0.051	11

Table 3. *Mean ± standard deviation of per-category accuracy across the 11 BBQ bias categories, by context condition. Ambiguous-context accuracy drops sharply C1→C3 on both models; disambiguating-context accuracy is stable.*

The ambiguous-disambiguating split in Table 3 mirrors the bootstrap findings of Table 2 but at the category-aggregated level. Disambiguating-context accuracy is stable across all three conditions on both models (0.752 → 0.761 → 0.798 for the 2B model; 0.926 → 0.943 → 0.931 for the 7B model). The entire accuracy drop is borne by ambiguous-context items. The per-category standard deviation in the ambiguous condition is 0.07–0.10, indicating that the effect is broadly distributed rather than driven by one outlier category.

5.6 The sAMB rise is accuracy-driven: premature commitment under visual input

Figure 3 shows sAMB rising monotonically in most categories C1→C3, which on a first reading suggests that images are making the model more stereotype-aligned. However, sAMB is a product of two quantities (§3.2): the ambiguous-context commitment error rate (1 – Accuracy_{amb}) and the raw stereotype-commitment ratio on ambiguous items (sDIS_{amb}). To distinguish whether the observed sAMB rise reflects more committing or more stereotyped committing, we compute both components separately. Table 4 shows them side-by-side for all six (model × condition) cells.

Model	Cond.	Acc. (amb)	1 – Acc	sDIS_amb	sAMB	n_commit
Qwen2-VL-2B	C1	0.310	0.690	+0.115	+0.079	20,187
	C2	0.255	0.745	+0.102	+0.076	21,781
	C3	0.169	0.831	+0.101	+0.084	24,101
Qwen2-VL-7B	C1	0.668	0.332	+0.295	+0.098	9,722
	C2	0.443	0.557	+0.197	+0.110	16,298
	C3	0.369	0.631	+0.179	+0.113	18,310

Table 4. *sAMB decomposition on all 11 BBQ categories pooled. sDIS_{amb} is the raw stereotype-commitment ratio $2\cdot(n_{biased_amb} / n_{non\text{-}unknown_amb}) - 1$ computed on ambiguous items only; n_commit is the number of ambiguous items the model committed to a named person (i.e. chose a non-unknown answer). On Qwen2-VL-7B, sDIS_{amb} falls from +0.295 (C1) to +0.179 (C3); the sAMB rise from +0.098 to +0.113 is entirely explained by the (1 – Accuracy) term rising from 0.33 to 0.63. Commitments on ambiguous items nearly double (9,722 → 18,310). The same qualitative pattern holds for Qwen2-VL-2B.*

The mechanism is therefore not stereotype amplification but premature commitment under visual input. On Qwen2-VL-7B, adding a real SDXL image nearly doubles the number of ambiguous items on which the model abandons the "unknown" answer in favour of a named person (9,722 → 18,310). The bias ratio at which those additional commitments occur (sDIS_{amb}) is *lower* than the text-only baseline, not higher. Images do not make the model pick more stereotyped answers from among the named options; they make the model pick a named option at all, when the correct action is to refuse to commit.

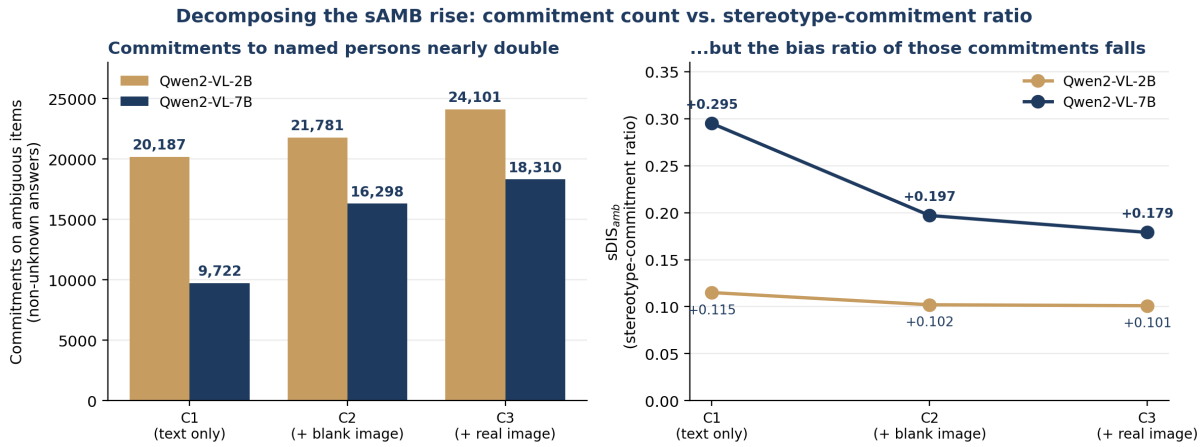

Figure 4. *The sAMB decomposition visualized.* ***Left:*** *commitments to named persons on ambiguous items rise sharply C1→C3 on both models (Qwen2-VL-7B: 9,722 → 18,310, nearly 2×).* ***Right:*** *the stereotype-commitment ratio sDIS_{amb} falls over the same conditions (7B: +0.295 → +0.179). The rise in sAMB across conditions is therefore a multiplicative artefact of the rising commitment rate, not a symptom of more stereotype-aligned commitments.*

The per-category version of this decomposition (11 categories × 3 conditions for Qwen2-VL-7B) confirms the pattern: sDIS_{amb} drifts C1→C3 are negative in 9 of 11 categories, including the high-sAMB categories

of Physical appearance (–0.107), Age (–0.071), Gender identity (–0.168), Religion (–0.147), and Race×gender (–0.139). Only Disability status (+0.052) and Race/ethnicity (+0.038) show small positive drifts; these two categories have the smallest denominators and the drifts are within the range we would attribute to commitment-count sampling noise. Full per-category $sDIS_{amb}$ values are tabulated in *results/analysis/bias_scores.csv* in the accompanying repository.

6. Discussion

6.1 Model size and the architecture-mode penalty

The total visual accuracy drop is substantially larger on the 7B model than on the 2B model (–14.8 pp vs. –4.8 pp), which is counterintuitive: the 7B model has a higher text-only baseline and one might expect better text reasoning to be more robust to the addition of an image. Figure 2 resolves this. The 7B model's excess drop is driven almost entirely by its architecture-mode penalty (10.6 pp, vs. 2.2 pp on the 2B model), not by differential image-content sensitivity (4.2 pp vs. 2.6 pp). The 7B model's problem is therefore not that it "trusts the image too much" — it is that simply running its vision branch with *any* input degrades its text reasoning, and that degradation scales with model size.

A plausible mechanism is that the M-RoPE positional recomputation and the insertion of ~250 near-uniform visual tokens into the attention stream dilute the attention paid to text tokens, and that this dilution costs more in models with stronger text reasoning. Alternatively, the 7B model's larger LLM backbone may be more sensitive to mild out-of-distribution shifts in its input sequence statistics. Distinguishing these hypotheses would require ablations beyond the scope of the present study.

6.2 Ambiguity as the trigger for premature commitment

The near-null C3 effect in disambiguating contexts (7B diff = +0.002, $p = 0.083$) is our cleanest positive result, because it cleanly separates the conditions under which the image changes model behaviour from those under which it does not. When the text provides sufficient evidence to identify the correct individual, the image is effectively ignored. When the text leaves the identity ambiguous, the image changes behaviour dramatically — but §5.6 shows that the change is *not* a shift toward more stereotype-aligned commitments.

A more accurate framing is as follows. In ambiguous contexts the correct action is to answer "unknown". In text-only mode, the 7B model does this correctly about two thirds of the time ($Accuracy_{amb} = 0.668$). When the visual branch is activated, the model commits to a named person far more often: the number of ambiguous items on which the model chose a non-unknown answer rises from 9,722 (C1) to 18,310 (C3). These additional commitments are not *more* stereotype-aligned than the baseline commitments — $sDIS_{amb}$ in fact drops from +0.295 to +0.179 — but since the baseline commitment ratio is well above zero, a larger number of commitments at a comparable bias ratio still produces a larger absolute count of biased errors. The visual branch, in effect, breaks the model's willingness to say "I do not have enough information."

6.3 Why the per-category pattern matters

Figure 3 shows that the sAMB rise C1→C3 concentrates in visually inferable categories: Physical appearance, Age, Disability status, and Gender identity. If the residual C2→C3 effect were simply additional architecture-mode noise from more complex visual patches, it would have no reason to distribute unevenly across categories according to how visually inferable the underlying social dimension is. The per-category concentration is therefore real evidence that the image content — not merely the act of processing any image — is driving the C2→C3 residual.

However, §5.6 shows that this concentration operates through commitment counts, not through the bias ratio of the commitments that result. The 7B model nearly doubles its commitment rate on Physical appearance items (C1: 297; C3: 544) and on Gender identity items (C1: 813; C3: 1,885), because the image provides visually-salient cues (apparent age, apparent gender, apparent physical features) that the model treats as evidence warranting commitment to a named person. On Nationality — where apparent visual cues are weak indicators of actual nationality — the commitment-count increase is much smaller, and the sAMB barely drifts. The per-category pattern thus reflects how readily each bias dimension admits a visual proxy, which is precisely the content-driven signature the original framing intended to capture — but the mechanism by which that signature produces biased outputs is epistemic overconfidence in response to image content, not stereotype-aligned answer selection.

6.4 Limitations

Four caveats deserve explicit mention. First, 4-bit NF4 quantisation may amplify noise in sensitive comparisons; full-precision inference (not feasible on the 24 GB RTX 4090 used here, particularly for 7B) could yield slightly different effect sizes. Second, 0.82% of SDXL generations produced invalid outputs and were excluded from C3; this is documented per condition in the data-coverage section of the accompanying *summary_report.txt*, and the paired-intersection design means those 481 rows are excluded from all C3-involved comparisons in a principled way. Third, only two model sizes from the Qwen2-VL family are tested; the architecture-mode vs. content-effect decomposition, and the premature-commitment interpretation, should be verified on other LVLM families (LLaVA, PaliGemma, Gemma 3, InternVL) before the quantitative split is treated as general. Fourth, the sAMB decomposition in §5.6 attributes the sAMB rise to the commitment-rate component of sAMB and not to the stereotype-ratio component. This interpretation depends on the specific multiplicative structure of the BBQ bias score as defined by Parrish et al.; alternative scoring conventions (e.g. unscaled stereotype-target rates, or scoring that does not treat “unknown” as a first-class answer option) would produce a different decomposition and potentially a different narrative.

7. Conclusion

This paper reports two decompositions of what prior work has termed *visual over-reliance* in Large Vision-Language Models. The first, enabled by the three-condition design, separates the total visual effect into an architecture-mode penalty (C1→C2) and an image-content effect (C2→C3). On Qwen2-VL-7B, the architecture-mode penalty accounts for ~72% of the total accuracy drop, incurred simply by activating the vision encoder with any content. The image-content effect accounts for the remaining ~28%, and is ambiguity-specific: disambiguating-context performance is unaffected by image content.

The second decomposition factors the ambiguous-context bias score sAMB into its raw stereotype-commitment ratio ($sDIS_{amb}$) and its (1 − Accuracy) multiplier. This yields the paper's main qualitative finding: the sAMB rise across conditions — concentrated in visually inferable categories, and superficially suggestive of stereotype amplification — is driven by the accuracy component, not the stereotype component. In 9 of 11 bias categories on Qwen2-VL-7B, $sDIS_{amb}$ is flat or declines from C1 to C3. Images do not make the model pick more stereotyped answers; they make the model pick a named answer at all when the correct response is to withhold judgment. We name this mechanism *premature commitment under visual input*.

The deployment implications follow from the decompositions. The architecture-mode penalty suggests prompting strategies or routing policies that make the vision-encoder path optional when the task is primarily textual. The premature-commitment effect suggests that image-conditioned abstention is a more promising intervention target than stereotype-specific answer reweighting: the model's committed answers are no more biased with images than without; it is the *willingness to commit* that shifts. Existing debiasing

techniques targeted at answer distribution [Jung et al. 2024; Pang et al. 2025] address a symptom that, on our data, is not the primary driver of the observed effect.

Full code, generated images, prediction CSVs, and analysis outputs are available at: https://github.com/logvisionn/bbq-visual-overreliance.

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